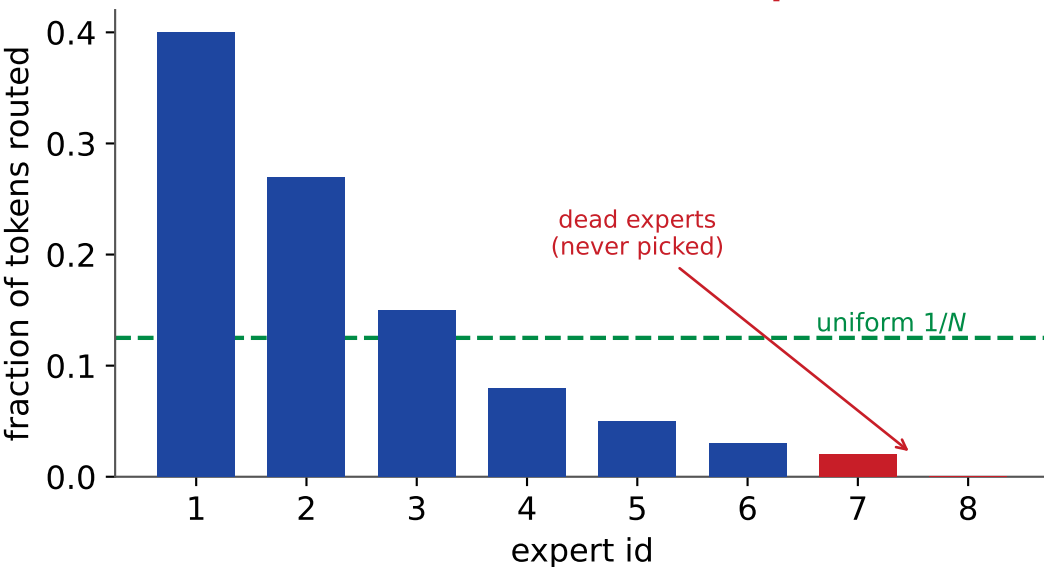


Load-balancing auxiliary loss keeps all experts used (schematic;  $N = 8$ )

**Without aux loss: collapse**



**With aux loss  $\alpha N \sum_i f_i P_i$  : balanced**

